

Code No: 183AH**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech II Year I Semester Examinations, February - 2024****COMPUTER ORGANIZATION AND ARCHITECTURE****(Common to CSE(AI&ML), AI&DS, AI&ML)****Time: 3 Hours****Max. Marks: 60****Note:** This question paper contains two parts A and B.i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A**(10 Marks)**

- 1.a) What is a register transfer language? [1]
- b) Define computer registers. [1]
- c) What is Micro-programmed control Unit? [1]
- d) List any two types of logical and bit manipulation instructions. [1]
- e) What is the purpose of data representation in computers? [1]
- f) Give an example of a real-world application that heavily relies on floating-point arithmetic. [1]
- g) What is the purpose of data types in computer programming? [1]
- h) Why is cache memory faster than main memory? [1]
- i) List the key features of an array processor. [1]
- j) Define the term "RISC". [1]

PART-B**(50 Marks)**

- 2.a) Give a detailed overview of digital computers, including their key characteristics and applications.
 - b) Compare computer design with computer architecture. [6+4]
- OR**
- 3.a) Describe the phases of instruction cycle.
 - b) Discuss about arithmetic micro operations. [5+5]
4. Explain the following terms.
 - a) Design of control unit.
 - b) Micro program example. [5+5]
- OR**
5. Explain various addressing modes with examples. [10]
 6. Justify the inclusion of a separate decimal arithmetic unit in modern computer architectures, citing real-world applications. [10]

OR

- 7.a) Discuss the challenges associated with decimal arithmetic operations.
b) Discuss any common multiplication algorithm used in computer arithmetic. [5+5]

- 8.a) Differentiate between Main Memory and Auxiliary Memory.
b) Explain the importance of memory hierarchy and its importance in data storage. [4+6]

OR

- 9.a) Write short notes on programmed I/O mode of transfer.
b) Explain importance of the I/O Interface in ensuring effective communication between a computer system and external devices. [5+5]

- 10.a) Explain about Inter-processor communication and synchronization in multiprocessors.
b) Differentiate between arithmetic pipeline and instruction pipeline. [5+5]

OR

- 11.a) State and explain the characteristics of multi-processors.
b) Explain RISC characteristics with its advantages [5+5]

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